

LAB ACTIVITY (Exp-01)

1. Maximum Subarray Sum - Kadane's Algorithm

Given an integer array arr[], find the subarray (containing at least one element) which has the maximum possible sum, and return that sum

Brute Force Method:-

```
#include <stdio.h>
#include <limits.h>

void findMaxSubarrayBruteForce(int arr[], int n) {
    int maxSum = INT_MIN;
    int startIdx = -1;
    int endIdx = -1;

    for (int i = 0; i < n; i++) {
        int currentSum = 0;

        for (int j = i; j < n; j++) {
            currentSum += arr[j];

            if (currentSum > maxSum) {
                maxSum = currentSum;
                startIdx = i;
                endIdx = j;
            }
        }
    }

    printf("Maximum Subarray Sum: %d\n", maxSum);
    printf("Subarray: [");
    for (int k = startIdx; k <= endIdx; k++) {
        printf("%d", arr[k]);
        if (k < endIdx) {
            printf(", ");
        }
    }
    printf("]\n");
}

int main() {
    int n;
    printf("Enter the length of array: ");
    scanf("%d", &n);

    int arr[n];
    printf("Enter the values of array");
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
}
```

```

    findMaxSubarrayBruteForce(arr, n);

    return 0;
}

```

```

PS D:\SEMESTER I\LAB WORK\Data structures and algorithm\lab5(2409)> cd "Data structures and algorithm\lab1(0608)" ; if ($?) { gcc maxsubarray.c -o maxsubarray }
Enter the length of array: 7
Enter the values of array: 2 3 -8 7 -1 2 3
Maximum Subarray Sum: 11
Subarray: [7, -1, 2, 3]
PS D:\SEMESTER I\LAB WORK\Data structures and algorithm\lab1(0608)> cd "Data structures and algorithm\lab1(0608)" ; if ($?) { gcc maxsubarray.c -o maxsubarray }
Enter the length of array: 2
Enter the values of array: -2 -4
Maximum Subarray Sum: -2
Subarray: [-2]
PS D:\SEMESTER I\LAB WORK\Data structures and algorithm\lab1(0608)> cd "Data structures and algorithm\lab1(0608)" ; if ($?) { gcc maxsubarray.c -o maxsubarray }
Enter the length of array: 5
Enter the values of array: 5 4 1 7 8
Maximum Subarray Sum: 25
Subarray: [5, 4, 1, 7, 8]
PS D:\SEMESTER I\LAB WORK\Data structures and algorithm\lab1(0608)>

```

Dynamic Method:-

```

#include<stdio.h>

void main() {
    int n;
    printf("Enter the length of array: ");
    scanf("%d",&n);

    int arr[n],i;
    printf("Enter the values of array");
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
}

```

```

int max_c= arr[0];
int max_g=arr[0];
int start=0,end=0,s=0;
for(i=1;i<n;i++){
    if(arr[i]>(max_c+arr[i])){
        max_c=arr[i];
        s=i;
    }
    else
        max_c=max_c+arr[i];
    if(max_c>max_g){
        max_g=max_c;
        start=s;
        end=i;
    }
}
printf("Maximum Subarray Sum: %d\n", max_g);
printf("Subarray: [");
for (int k = start; k <= end; k++) {
    printf("%d", arr[k]);
    if (k < end) {
        printf(", ");
    }
}
printf("]\n");
}

```

```

PS D:\SEMESTER I\LAB WORK\Data structures and algorithm\lab1(0608)> cd "
tructures and algorithm\lab1(0608)" ; if ($?) { gcc kadane.c -o kadane
Enter the length of array: 7
Enter the values of array 2 3 -8 7 -1 2 3
Maximum Subarray Sum: 11
Subarray: [7, -1, 2, 3]
PS D:\SEMESTER I\LAB WORK\Data structures and algorithm\lab1(0608)> cd "
tructures and algorithm\lab1(0608)" ; if ($?) { gcc kadane.c -o kadane
Enter the length of array: 2
Enter the values of array -2 -4
Maximum Subarray Sum: -2
Subarray: [-2]
PS D:\SEMESTER I\LAB WORK\Data structures and algorithm\lab1(0608)> cd "
tructures and algorithm\lab1(0608)" ; if ($?) { gcc kadane.c -o kadane
Enter the length of array: 5
Enter the values of array 5 4 1 7 8
Maximum Subarray Sum: 25
Subarray: [5, 4, 1, 7, 8]
PS D:\SEMESTER I\LAB WORK\Data structures and algorithm\lab1(0608)>

```

2. Maximum consecutive ones after flipping zeroes

- Input Array
- Constant value=k (the no. of zeroes you can flip)

```
#include<stdio.h>
#include<stdlib.h>

void main(){
    int n;
    printf("Enter the length of array: ");
    scanf("%d",&n);

    int arr[n],i;
    printf("Enter the values of array");
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int k;
    printf("Enter the value of k: ");
    scanf("%d",&k);

    int left = 0;
    int right = 0;
    int zeroCount = 0;
    int maxLen = 0;

    while (right < n) {
        if (arr[right] == 0) {
            zeroCount++;
        }

        while (zeroCount > k) {
            if (arr[left] == 0) {
                zeroCount--;
            }
            left++;
        }

        int currentWindowLength = right - left + 1;
        if (currentWindowLength > maxLen) {
            maxLen = currentWindowLength;
        }

        right++;
    }

    printf("Maximum consecutive ones after flipping at most %d zeros: %d\n", k, maxLen);
}
```

```
PS D:\SEMESTER I\LAB WORK\Data structures and algorithm\lab1(0608)> cd "d:\Data structures and algorithm\lab2(1308)\\" ; if ($?) { gcc maxconsecones.c -o maxconsecones }
Enter the length of array: 11
Enter the values of array 1 1 1 0 0 0 1 1 1 1 0
Enter the value of k: 2
Maximum consecutive ones after flipping at most 2 zeros: 6
PS D:\SEMESTER I\LAB WORK\Data structures and algorithm\lab2(1308)> cd "d:\Data structures and algorithm\lab2(1308)\\" ; if ($?) { gcc maxconsecones.c -o maxconsecones }
Enter the length of array: 7
Enter the values of array 1 0 0 1 0 1 1
Enter the value of k: 2
Maximum consecutive ones after flipping at most 2 zeros: 5
PS D:\SEMESTER I\LAB WORK\Data structures and algorithm\lab2(1308)> |
```